

BLM2041 Signals and Systems

Week 5

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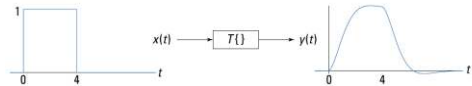
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Responses to arbitrary signals

- Although we have focused on responses to simple signals ($\delta[n]$, $\delta(t)$) we are generally interested in responses to more complicated signals.
- How do we compute responses to a more complicated input signals?



Block diagram depicting a general input/output relationship.

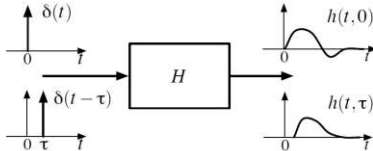
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Impulse Response

The *impulse response* of a linear system $h_\tau(t)$ is the output of the system at time t to an impulse at time τ . This can be written as

$$h_\tau = H(\delta_\tau)$$

Care is required in interpreting this expression!



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Note: Be aware of potential confusion here:

When you write

$$h_\tau(t) = H(\delta_\tau(t))$$

the variable t serves different roles on each side of the equation.

- t on the left is a specific value for time, the time at which the output is being sampled.
- t on the right is varying over all real numbers, it is not the same t as on the left.
- The output at time specific time t on the left in general depends on the input at all times t on the right (the entire input waveform).

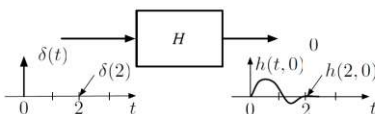
- Assume the input impulse is at $\tau = 0$,

$$h = h_0 = H(\delta_0).$$

We want to know the impulse response at time $t = 2$. It doesn't make any sense to set $t = 2$, and write

$$h(2) = H(\delta(2)) \quad \Leftarrow \text{No!}$$

First, $\delta(2)$ is something like zero, so $H(0)$ would be zero. Second, the value of $h(2)$ depends on the entire input waveform, not just the value at $t = 2$.

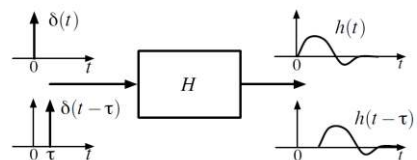


Time-invariance

If H is time invariant, delaying the input and output both by a time τ should produce the same response

$$h_\tau(t) = h(t - \tau).$$

In this case, we don't need to worry about h_τ because it is just h shifted in time.



Linearity and Extended Linearity

Linearity: A system S is linear if it satisfies both

- **Homogeneity:** If $y = Sx$, and a is a constant then

$$ay = S(ax).$$

- **Superposition:** If $y_1 = Sx_1$ and $y_2 = Sx_2$, then

$$y_1 + y_2 = S(x_1 + x_2).$$

Combined Homogeneity and Superposition:

If $y_1 = Sx_1$ and $y_2 = Sx_2$, and a and b are constants,

$$ay_1 + by_2 = S(ax_1 + bx_2)$$

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Extended Linearity

- **Summation:** If $y_n = S(x_n)$ for all n , an integer from $(-\infty < n < \infty)$, and a_n are constants

$$\sum_n a_n y_n = S\left(\sum_n a_n x_n\right)$$

Summation and the system operator commute, and can be interchanged.

- **Integration (Simple Example):** If $y = S(x)$,

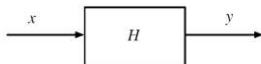
$$\int_{-\infty}^{\infty} a(\tau) y(t - \tau) d\tau = S\left(\int_{-\infty}^{\infty} a(\tau) x(t - \tau) d\tau\right)$$

Integration and the system operator commute, and can be interchanged.

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Output of an LTI System

We would like to determine an expression for the output $y(t)$ of an linear time invariant system, given an input $x(t)$



We can write a signal $x(t)$ as a sample of itself

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta_{\tau}(t) d\tau$$

This means that $x(t)$ can be written as a weighted integral of δ functions.

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Applying the system H to the input $x(t)$,

$$\begin{aligned} y(t) &= H(x(t)) \\ &= H\left(\int_{-\infty}^{\infty} x(\tau) \delta_{\tau}(t) d\tau\right) \end{aligned}$$

If the system obeys extended linearity we can interchange the order of the system operator and the integration

$$y(t) = \int_{-\infty}^{\infty} x(\tau) H(\delta_{\tau}(t)) d\tau.$$

The impulse response is

$$h_{\tau}(t) = H(\delta_{\tau}(t)).$$

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Substituting for the impulse response gives

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h_{\tau}(t) d\tau.$$

This is a *superposition integral*. The values of $x(\tau)h(t, \tau)d\tau$ are superimposed (added up) for each input time τ .

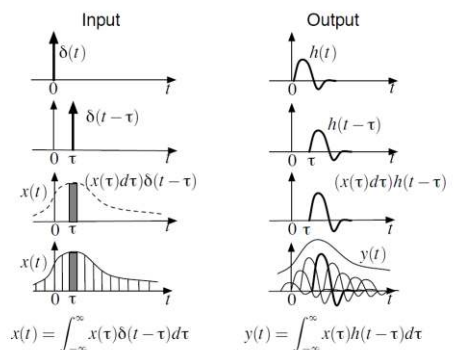
If H is time invariant, this written more simply as

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h_{\tau}(t) d\tau.$$

This is in the form of a *convolution integral*, which will be the subject of the next class.

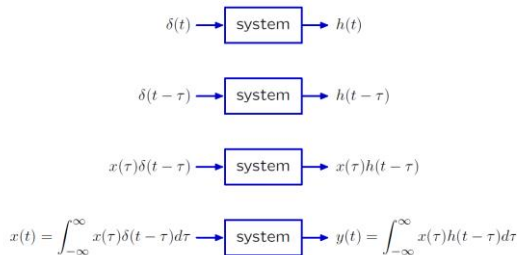
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Graphically, this can be represented as:



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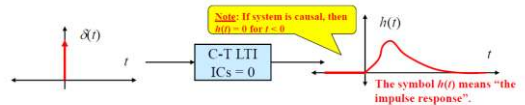
If a system is linear and time-invariant (LTI) then its output is the integral of weighted and shifted unit-impulse responses.



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Recall: Impulse Response

Earlier we introduced the concept of impulse response...
...what comes out of a system when the input is an impulse (delta function)



Noting that the LT of $\delta(t) = 1$ and using the properties of the transfer function and the Z transform we said that

$$h(t) = \mathcal{L}^{-1}\{H(s)\mathcal{L}\{\delta(t)\}\} \quad h(t) = \mathcal{L}^{-1}\{H(s)\} \quad h(t) = \mathcal{Z}^{-1}\{H(\omega)\}$$

So...once we have either $H(s)$ or $H(\omega)$ we can get the impulse response $h(t)$

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Convolution Property and System Output

Let $x(t)$ be a signal with CTFT $X(\omega)$ and LT of $X(s)$

Consider a system w/ freq resp $H(\omega)$ & trans func $H(s)$

$$\begin{aligned} x(t) &\leftrightarrow X(\omega) \\ x(t) &\leftrightarrow X(s) \\ h(t) &\leftrightarrow H(\omega) \\ h(t) &\leftrightarrow H(s) \end{aligned}$$

We've spent much time using these tools to analyze system outputs this way:

$$Y(\omega) = H(\omega)X(\omega) \leftrightarrow y(t) = \mathcal{F}^{-1}\{H(\omega)X(\omega)\}$$

$$Y(s) = H(s)X(s) \leftrightarrow y(t) = \mathcal{L}^{-1}\{H(s)X(s)\}$$

The convolution property of the CTFT and LT gives an alternate way to find $y(t)$:

$$\begin{aligned} \mathcal{F}^{-1}\{X(\omega)H(\omega)\} &= x(t) * h(t) \\ \mathcal{L}^{-1}\{X(s)H(s)\} &= x(t) * h(t) \\ x(t) * h(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \\ y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \end{aligned}$$

"Convolution" input $x(t)$ with the impulse response $h(t)$ gives the output $y(t)$!

LTI System with impulse response $h(t)$

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Convolution for Causal System & with Causal Input

An arbitrary LTI system's output can be found using the general convolution form:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \quad \text{General LTI System}$$

If the system is causal then $h(t) = 0$ for $t < 0$... Thus $h(t-\tau) = 0$ for $t > \tau$... so:

$$y(t) = \int_{-\infty}^t x(\tau)h(t-\tau)d\tau \quad \text{Causal LTI System}$$

If the input is causal then $x(t) = 0$ for $t < 0$... so:

$$y(t) = \int_0^t x(\tau)h(t-\tau)d\tau \quad \text{Causal Input & General LTI System}$$

If the system & signal are both causal then

$$y(t) = \int_0^t x(\tau)h(t-\tau)d\tau \quad \text{Causal Input & Causal LTI System}$$

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Convolution Properties

1. **Commutativity** $x(t) * h(t) = h(t) * x(t)$

2. **Associativity** $[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$

Associativity together with commutativity says we **can interchange the order of two cascaded systems**.

3. **Distributivity** $x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$

4. **Derivative Property:** $\frac{d}{dt}[x(t) * v(t)] = \dot{x}(t) * v(t) = x(t) * \dot{v}(t)$ derivative

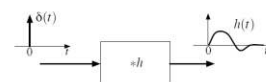
5. **Integration Property** Let $y(t) = x(t) * h(t)$, then

$$\int_{-\infty}^t y(\lambda)d\lambda = \left[\int_{-\infty}^t x(\lambda)d\lambda \right] * h(t) = x(t) * \left[\int_{-\infty}^t h(\lambda)d\lambda \right]$$

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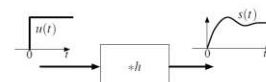
Example: Measuring the impulse response of an LTI system.

We would like to measure the impulse response of an LTI system, described by the impulse response $h(t)$



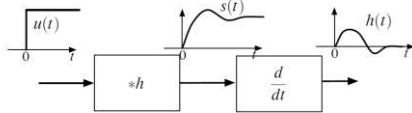
This can be practically difficult because input amplitude is often limited. A very short pulse then has very little energy.

A common alternative is to measure the **step response** $s(t)$, the response to a unit step input $u(t)$

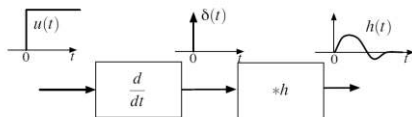


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The impulse response is determined by differentiating the step response,



To show this, commute the convolution system and the differentiator to produce a system with the same overall impulse response



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Steps for Graphical Convolution $x(t) * h(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

1. **Re-Write the signals as functions of τ :** $x(\tau)$ and $h(\tau)$
2. **Flip** just one of the signals around $t = 0$ to get either $x(-\tau)$ or $h(-\tau)$
 - a. It is usually best to flip the signal with shorter duration
 - b. For notational purposes here, we'll flip $h(\tau)$ to get $h(-\tau)$
3. **Find Edges** of the flipped signal
 - a. Find the left-hand-edge τ -value of $h(-\tau)$: **call it $\tau_{L,B}$**
 - b. Find the right-hand-edge τ -value of $h(-\tau)$: **call it $\tau_{R,B}$**
4. **Shift** $h(-\tau)$ by an arbitrary value of t to get $h(t-\tau)$ and **get its edges**
 - a. Find the left-hand-edge τ -value of $h(t-\tau)$ as a function of t : **call it $\tau_{L,t}$**
 - **Important:** It will always be... **$\tau_{L,t} = t + \tau_{L,B}$**
 - b. Find the right-hand-edge τ -value of $h(t-\tau)$ as a function of t : **call it $\tau_{R,t}$**
 - **Important:** It will always be... **$\tau_{R,t} = t + \tau_{R,B}$**

Note: If the signal you flipped is NOT finite duration, one or both of $\tau_{L,t}$ and $\tau_{R,t}$ will be infinite ($\tau_{L,t} = -\infty$ and/or $\tau_{R,t} = \infty$)

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Steps Continued

5. **Find Regions of τ -Overlap**
 - a. What you are trying to do here is find intervals of t over which the product $x(\tau)h(t-\tau)$ has a single mathematical form in terms of τ
 - b. In each region find: Interval of t that makes the identified overlap happen
 - c. Working examples is the best way to learn how this is done

Tips: Regions should be contiguous with no gaps!!!
Don't worry about $<$ vs. $\leq$ etc.

6. For Each Region: **Form the Product $x(\tau)h(t-\tau)$ and Integrate**
 - a. Form product $x(\tau)h(t-\tau)$
 - b. **Find the Limits of Integration** by finding the interval of τ over which the product is nonzero
 - i. Found by seeing where the edges of $x(\tau)$ and $h(t-\tau)$ lie
 - ii. Recall that the edges of $h(t-\tau)$ are $\tau_{L,t}$ and $\tau_{R,t}$, which often depend on the value of t
 - So... the limits of integration may depend on t
 - c. **Integrate the product $x(\tau)h(t-\tau)$ over the limits found in 6b**
 - i. The result is generally a function of t , but is only valid for the interval of t found for the current region
 - ii. Think of the result as a "time-section" of the output $y(t)$

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Steps Continued

7. **"Assemble" the output** from the output time-sections for all the regions
 - a. Note: you do NOT add the sections together
 - b. You define the output "piecewise"
 - c. Finally, if possible, look for a way to write the output in a simpler form

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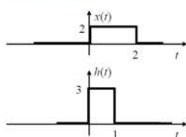
Example: Graphically Convolve Two Signals

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

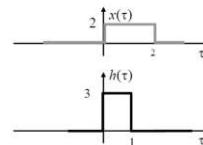
By "Properties of Convolution"... these two forms are Equal
This is why we can flip either signal

Convolve these two signals:

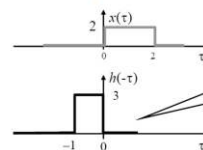


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Step #1: Write as Function of τ



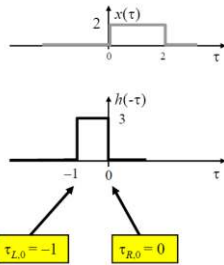
Step #2: Flip $h(\tau)$ to get $h(-\tau)$



Usually Easier to Flip the Shorter Signal

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Step #3: Find Edges of Flipped Signal

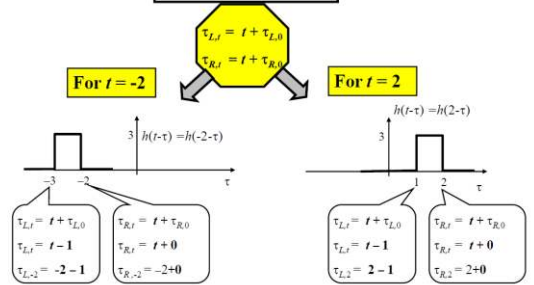


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Motivating Step #4: Shift by t to get $h(t-\tau)$ & Its Edges

Just looking at 2 "arbitrary" t values

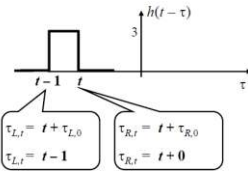
In Each Case We Get



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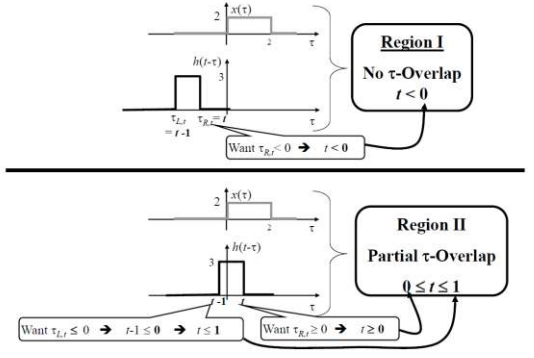
Doing Step #4: Shift by t to get $h(t-\tau)$ & Its Edges

For Arbitrary Shift by t



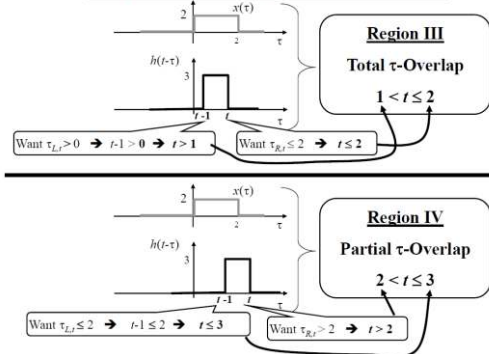
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Step #5: Find Regions of τ -Overlap



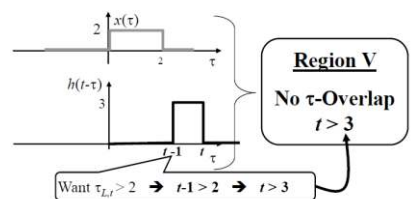
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Step #5 (Continued): Find Regions of τ -Overlap



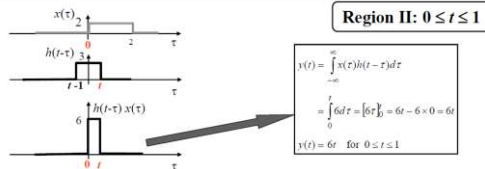
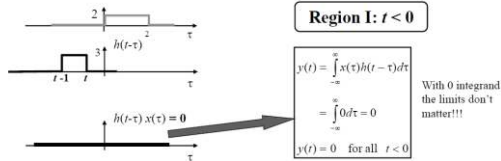
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Step #5 (Continued): Find Regions of τ -Overlap



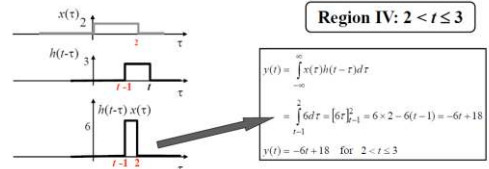
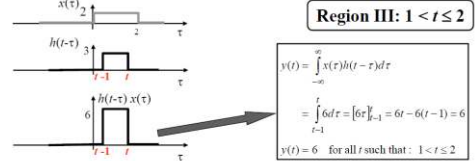
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Step #6: Form Product & Integrate For Each Region



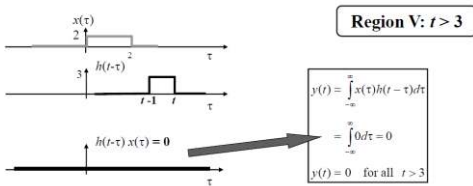
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Step #6 (Continued): Form Product & Integrate For Each Region



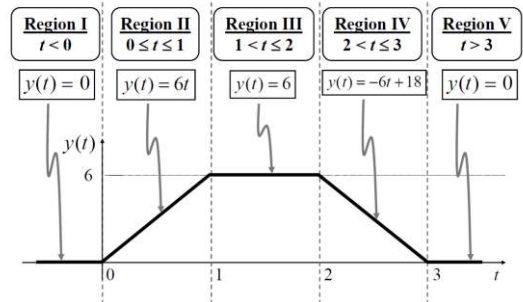
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Step #6 (Continued): Form Product & Integrate For Each Region



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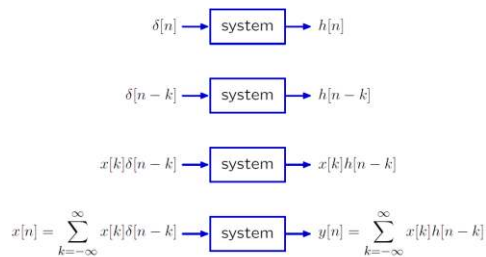
Step #7: "Assemble" Output Signal



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Discrete Convolution

If a system is linear and time-invariant (LTI) then its output is the sum of weighted and shifted unit-sample responses.

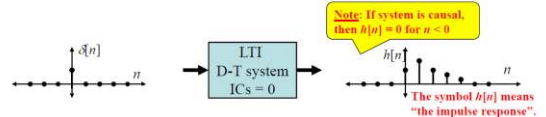


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Discrete Convolution

Recall: Impulse Response

Earlier we introduced the concept of impulse response...
...what comes out of a system when the input is an impulse (delta sequence)



Noting that the ZT of $\delta[n] = 1$ and using the properties of the transfer function and the Z transform we said that

$$h[n] = Z^{-1}\{H(z)Z\{\delta[n]\}\} \quad h[n] = Z^{-1}\{H(z)\} \quad h[n] = IDTFT\{H(\Omega)\}$$

So...once we have either $H(z)$ or $H(\Omega)$ we can get the impulse response $h[n]$

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Convolution Property and System Output

Let $x[n]$ be a signal with DTFT $X(\Omega)$ and ZT $X(z)$

$$x[n] \leftrightarrow X(\Omega)$$

$$x[n] \leftrightarrow X(z)$$

Consider a system w/ freq resp $H(\Omega)$ & trans func $H(z)$

$$h[n] \leftrightarrow H(\Omega)$$

$$h[n] \leftrightarrow H(z)$$

We've spent much time using these tools to analyze system outputs this way:

$$Y(\Omega) = H(\Omega)X(\Omega) \leftrightarrow y[n] = \text{DFT}^{-1}\{H(\Omega)X(\Omega)\}$$

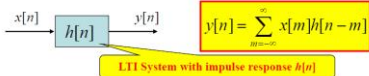
$$Y(z) = H(z)X(z) \leftrightarrow y[n] = \text{Z}^{-1}\{H(z)X(z)\}$$

The convolution property of the DFT and ZT gives an alternate way to find $y[n]$:

$$\text{DFT}^{-1}\{X(\Omega)H(\Omega)\} = x[n] * h[n]$$

$$x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m]$$

$$\text{Z}^{-1}\{X(z)H(z)\} = x[n] * h[n]$$



$$y[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m]$$

"Convolution" input $x[n]$ with the impulse response $h[n]$ gives the output $y[n]$!

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Convolution for Causal System & with Causal Input

An arbitrary LTI system's output can be found using the general convolution form:

$$y[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] \quad \text{General LTI System}$$

If the system is causal then $h[n] = 0$ for $n < 0 \dots$. Thus $h[n-m] = 0$ for $m > n \dots$ so:

$$y[n] = \sum_{m=0}^n x[m]h[n-m] \quad \text{Causal LTI System}$$

If the input is causal then $x[n] = 0$ for $n < 0 \dots$ so:

$$y[n] = \sum_{m=0}^{\infty} x[m]h[n-m] \quad \text{Causal Input & General LTI System}$$

If the system & signal are both causal then

$$y[n] = \sum_{m=0}^n x[m]h[n-m] \quad \text{Causal Input & Causal LTI System}$$

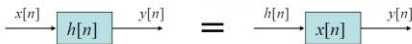
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Convolution Properties (can sometimes exploit to make things easier)

1. Commutativity

$$x[n] * h[n] = h[n] * x[n]$$

$$\sum_{m=-\infty}^{\infty} x[m]h[n-m] = \sum_{m=-\infty}^{\infty} h[m]x[n-m]$$



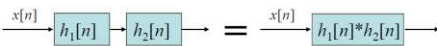
This is obvious from the frequency domain (or z domain) viewpoint:

$$x[n] * h[n] = h[n] * x[n] \Rightarrow X(\Omega)H(\Omega) = H(\Omega)X(\Omega)$$

2. Associativity

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$

$\Rightarrow$ Can combine cascade into single equivalent system

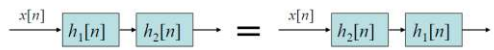


This is obvious from the frequency domain (or z domain) viewpoint:

$$[X(\Omega)H_1(\Omega)]H_2(\Omega) = X(\Omega)[H_1(\Omega)H_2(\Omega)] \quad \text{Tells us what the Freq Resp is for a cascade}$$

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Associativity together with commutativity says we can interchange the order of two cascaded systems:

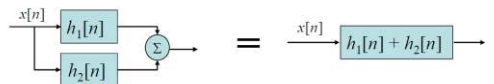


Warning: This holds in theory but in practice there may be physical issues that prevent this!!!

3. Distributivity

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

$\Rightarrow$ can combine sum of two outputs into a single system (or vice versa)



With commutativity this says we can split a complicated input into sum of simple ones... which is nothing more than "linearity"!!

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Graphical Convolution – To Visualize & Test Real Systems

Can do convolution this way when signals are known numerically or by equation

• Convolution involves the sum of a product of two signals: $x[i]h[n-i]$

• At each output index n , the product changes

Step 1: Write both as functions of i : $x[i]$ & $h[i]$

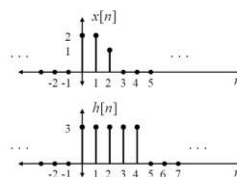
Step 2: Flip $h[i]$ to get $h[-i]$ (The book calls this "fold")

Repeat for each n **Step 3:** For each output index n value of interest, shift by n to get $h[n-i]$ (Note: positive n gives right shift!!!!)

Step 4: Form product $x[i]h[n-i]$ and sum its elements to get the number $y[n]$

"Commutativity" says we can flip either $x[i]$ or $h[i]$ and get the same answer

Example of Graphical Convolution



Find $y[n] = x[n] * h[n]$ for all integer values of n

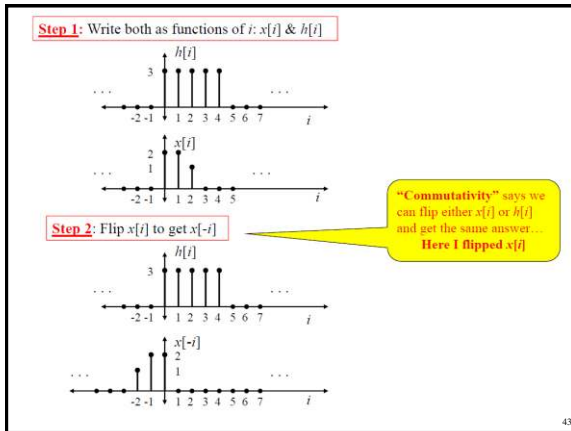
Solution

For this problem I choose to flip $x[n]$

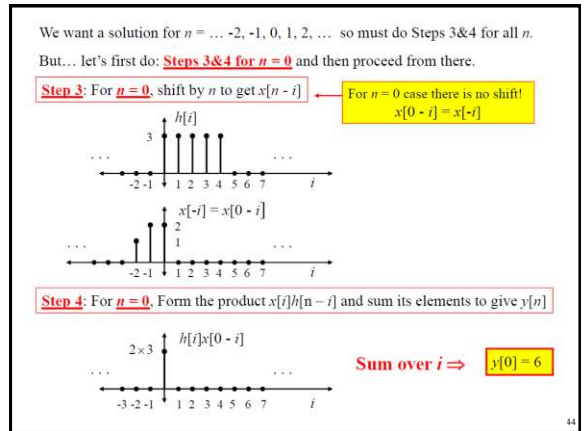
My personal preference is to flip the shorter signal although I sometimes don't follow that "rule"... only through lots of practice can you learn how to best choose which one to flip.

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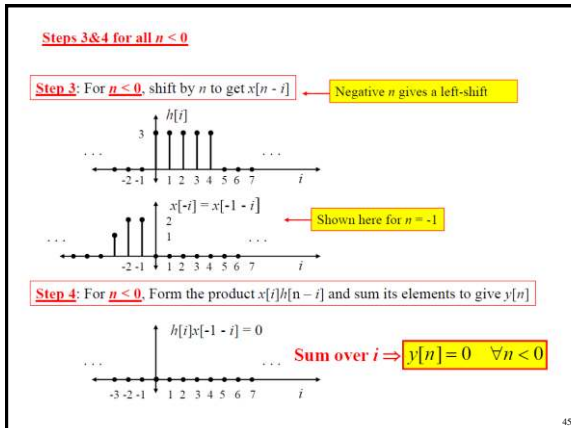
42



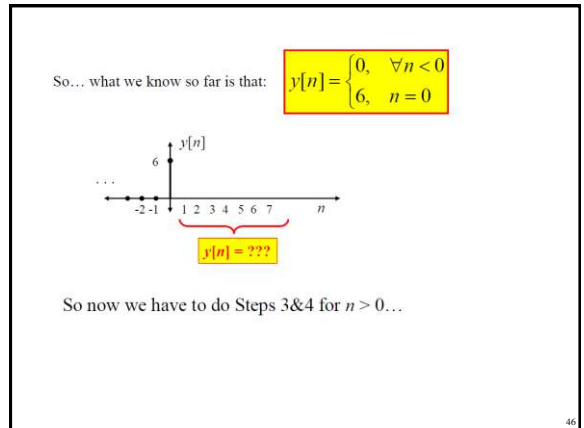
43



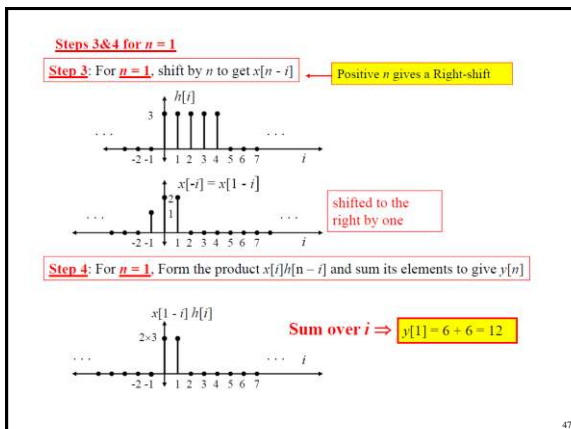
44



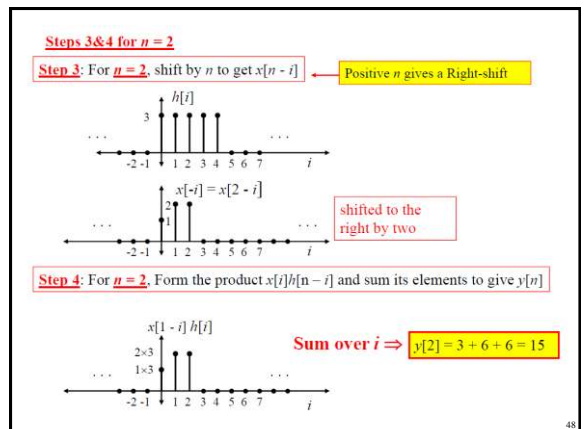
45



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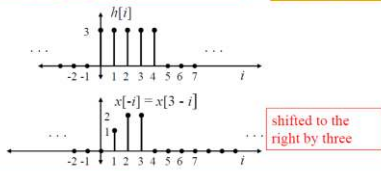
47



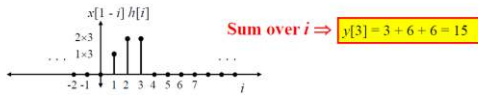
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Steps 3&4 for $n = 3$

Step 3: For $n = 3$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



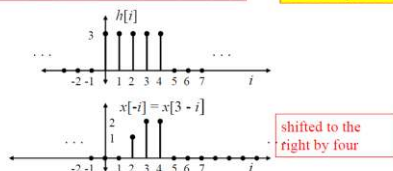
Step 4: For $n = 3$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



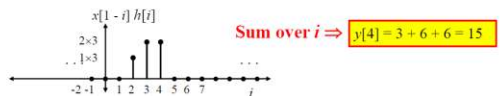
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Steps 3&4 for $n = 4$

Step 3: For $n = 4$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



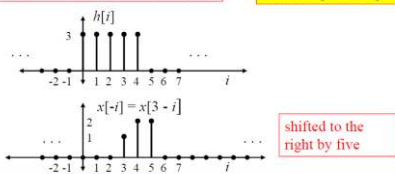
Step 4: For $n = 4$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



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Steps 3&4 for $n = 5$

Step 3: For $n = 5$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



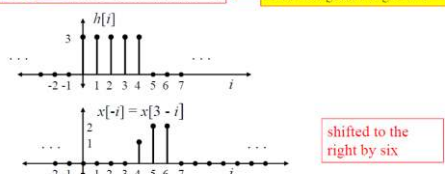
Step 4: For $n = 5$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



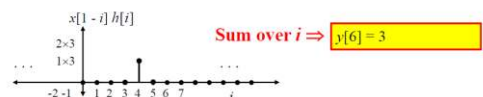
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Steps 3&4 for $n = 6$

Step 3: For $n = 6$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



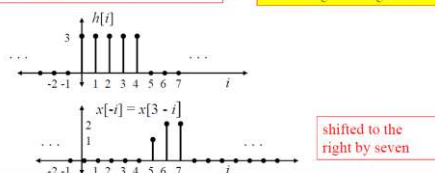
Step 4: For $n = 6$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$



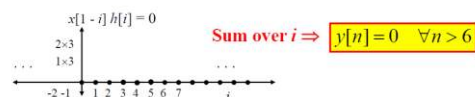
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Steps 3&4 for all $n > 6$

Step 3: For $n > 6$, shift by n to get $x[n - i]$ ← Positive n gives a Right-shift



Step 4: For $n > 6$, Form the product $x[i]h[n - i]$ and sum its elements to give $y[n]$

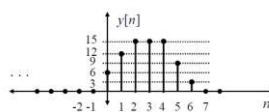


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So... now we know the values of $y[n]$ for all values of n

We just need to put it all together as a function...

Here it is easiest to just plot it... you could also list it as a table.



Note that convolving these kinds of signals gives a "ramp-up" at the beginning and a "ramp-down" at the end.

Various kinds of "transients" at the beginning and end of a convolution are common.

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